

H3CNE综合实验

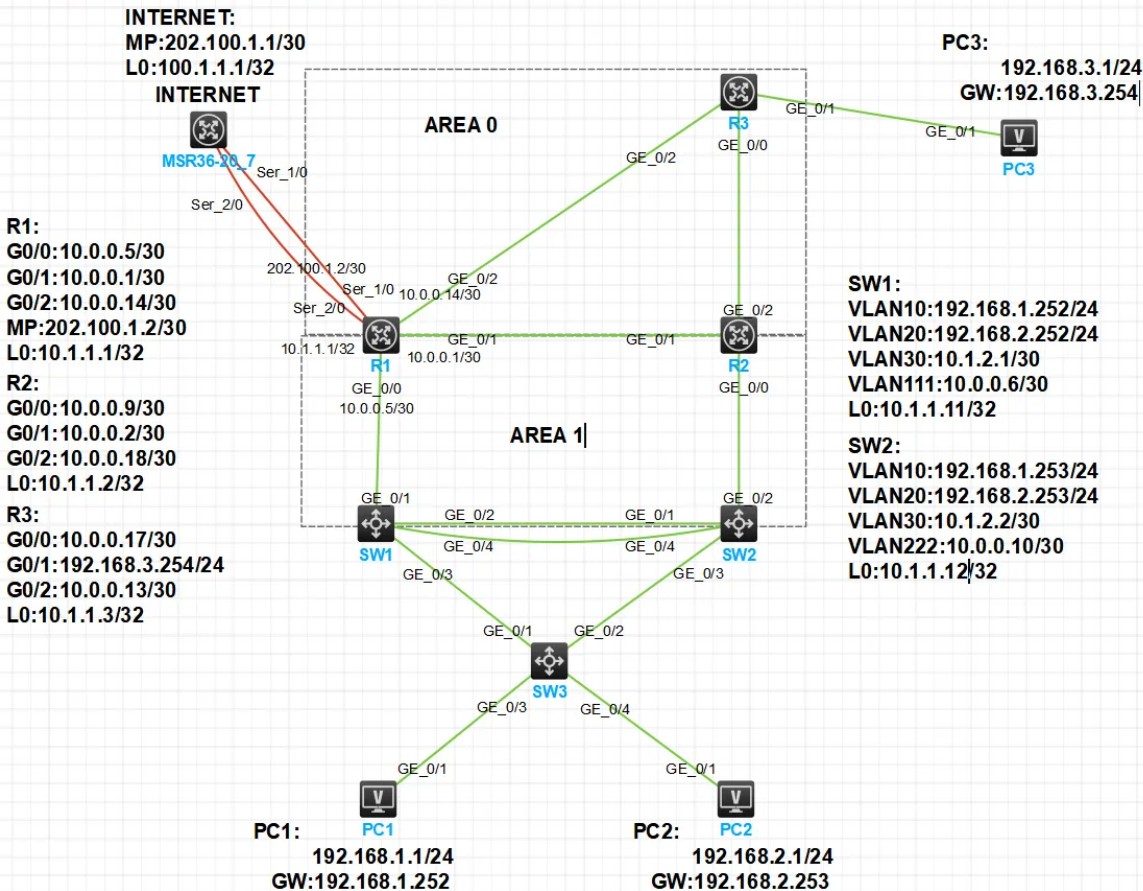
H3CNE综合实验

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实验拓扑

实验需求:

- 按照图示配置 IP 地址
- SW1 和 SW2 之间的直连链路配置链路聚合
- 公司内部业务网段为 Vlan10 和 Vlan20; Vlan10 是市场部, Vlan20 是技术部, 要求对 Vlan 进行命名以便识别;
PC1 属于 Vlan10, PC2 属于 Vlan20, Vlan30 用于 SW1 和 SW2 建立 OSPF 邻居; Vlan111 为 SW1 和 R1 的互联 Vlan,
Vlan222 为 SW2 和 R2 的互联 Vlan
- 所有交换机相连的端口配置为 Trunk, 允许相关流量通过
- 交换机连接 PC 的端口配置为边缘端口
- 在 SW1 上配置 DHCP 服务, 为 Vlan10 和 Vlan20 的 PC 动态分配 IP 地址、网关和 DNS 地址; 要求 Vlan10 的网关是 192.168.1.252, Vlan20 的网关是 192.168.2.253
- 按图示分区域配置 OSPF 实现公司内部网络全网互通, ABR 的环回口宣告进骨干区域; 业务网段不允许出现协议报文
- R1 上配置默认路由指向互联网, 并引入到 OSPF
- R1 通过双线连接到互联网, 配置 PPP-MP, 并配置双向 chap 验证
- 配置 EASY IP, 只有业务网段 192.168.1.0/24 和 192.168.2.0/24 的数据流可以通过 R1 访问互联网 11. R1 开启 TELNET 远程管理
使用用户 abc 登录, 密码 abc, 只允许技术部远程管理 R1



实验需求

1. 按照图示配置 IP 地址
2. SW1 和 SW2 之间的直连链路配置链路聚合
3. 公司内部业务网段为 Vlan10 和 Vlan20; Vlan10 是市场部, Vlan20 是技术部, 要求对 Vlan 进行命名以便识别; PC1 属于 Vlan10, PC2 属于 Vlan20, Vlan30 用于 SW1 和 SW2 建立 OSPF 邻居; Vlan111 为 SW1 和 R1 的互联 Vlan, Vlan222 为 SW2 和 R2 的互联 Vlan
4. 所有交换机相连的端口配置为 Trunk, 允许相关流量通过
5. 交换机连接 PC 的端口配置为边缘端口
6. 在 SW1 上配置 DHCP 服务, 为 Vlan10 和 Vlan20 的 PC 动态分配 IP 地址、网关和 DNS 地址; 要求 Vlan10 的网关是 192.168.1.252, Vlan20 的网关是 192.168.2.253
7. 按图所示区域配置 OSPF 实现公司内部网络全网互通, ABR 的环回口宣告进骨干区域; 业务网段不允许出现协议报文
8. R1 上配置默认路由指向互联网, 并引入到 OSPF
9. R1 通过双线连接到互联网, 配置 PPP-MP, 并配置双向 chap 验证
10. 配置 EASY IP, 只有业务网段 192.168.1.0/24 和 192.168.2.0/24 的数据流可以通过 R1 访问互联网
11. R1 开启 TELNET 远程管理, 使用用户 abc 登录, 密码 h3c1234567, 只允许技术部远程管理 R1

实验配置

1. 公司内部业务网段为 Vlan10 和 Vlan20; Vlan10 是市场部, Vlan20 是技术部, 要求对 Vlan 进行命名以便识别; PC1 属于 Vlan10, PC2 属于 Vlan20, Vlan30 用于 SW1 和 SW2 建立 OSPF 邻居; Vlan111 为 SW1 和 R1 的互联 Vlan, Vlan222 为 SW2 和 R2 的互联 Vlan

```
[SW1]vlan 10
[SW1-vlan10]name SC
[SW1-vlan10]vlan 20
[SW1-vlan20]name JS
[SW1-vlan20]vlan 30
[SW1]vlan 111
---
```

```
[SW2]vlan 10
[SW2-vlan10]name SC
[SW2-vlan10]VLAN 20
[SW2-vlan20]name JS
[SW2-vlan20]VLAN 30
[SW2]vlan 222
```

1. 所有交换机相连的端口配置为 Trunk, 允许相关流量通过

```
[SW1]vlan 10
[SW1-vlan10]vlan 20
[SW1-vlan20]vlan 30
```

```
[SW1]interface Bridge-Aggregation 1
[SW1]int g1/0/2
[SW1-GigabitEthernet1/0/2]port link-aggregation group 1
[SW1-GigabitEthernet1/0/2]int g1/0/4
[SW1-GigabitEthernet1/0/4]port link-aggregation group 1
[SW1-Bridge-Aggregation1]port link-type trunk
Configuring GigabitEthernet1/0/2 done.
Configuring GigabitEthernet1/0/4 done.
[SW1-Bridge-Aggregation1]port trunk permit vlan 10 20 30
Configuring GigabitEthernet1/0/2 done.
Configuring GigabitEthernet1/0/4 done.
[SW1-GigabitEthernet1/0/1]int g1/0/3
[SW1-GigabitEthernet1/0/3]port link-type trunk
[SW1-GigabitEthernet1/0/3]port trunk permit vlan 10 20
---
```

```
[SW2]vlan 10
[SW2-vlan10]vlan 20
[SW2-vlan20]vlan 30
[SW2]interface Bridge-Aggregation 1
[SW2]int g1/0/1
[SW2-GigabitEthernet1/0/1]port link-aggregation group 1
[SW2-GigabitEthernet1/0/1]int g1/0/4
[SW2-GigabitEthernet1/0/4]port link-aggregation group 1
[SW2]interface Bridge-Aggregation 1
[SW2-Bridge-Aggregation1]port link-type trunk
Configuring GigabitEthernet1/0/1 done.
Configuring GigabitEthernet1/0/4 done.
[SW2-Bridge-Aggregation1]port trunk permit vlan 10 20 30
Configuring GigabitEthernet1/0/1 done.
Configuring GigabitEthernet1/0/4 done.
[SW2-GigabitEthernet1/0/2]int g1/0/3
[SW2-GigabitEthernet1/0/3]port link-type trunk
[SW2-GigabitEthernet1/0/3]port trunk permit vlan 10 20
---
```

```
[SW3-GigabitEthernet1/0/3]int g1/0/1
[SW3-GigabitEthernet1/0/1]port link-type trunk
[SW3-GigabitEthernet1/0/1]port trunk permit vlan 10 20
[SW3-GigabitEthernet1/0/1]int g1/0/2
[SW3-GigabitEthernet1/0/2]port link-type trunk
[SW3-GigabitEthernet1/0/2]port trunk permit vlan 10 20
```

2. 按照图示配置 IP 地址

```
[R1]interface g0/0
[R1-GigabitEthernet0/0]ip ad 10.0.0.5 30
[R1-GigabitEthernet0/0]interface g0/1
[R1-GigabitEthernet0/1]ip ad 10.0.0.1 30
[R1-GigabitEthernet0/1]interface g0/2
[R1-GigabitEthernet0/2]ip ad 10.0.0.14 30
[R1-GigabitEthernet0/2]int l0
[R1-LoopBack0]ip ad 10.1.1.1 32
---
[R2]int g0/0
[R2-GigabitEthernet0/0]ip ad 10.0.0.9 30
```

```

[R2-GigabitEthernet0/0]int g0/1
[R2-GigabitEthernet0/1]ip ad 10.0.0.2 30
[R2-GigabitEthernet0/1]int g0/2
[R2-GigabitEthernet0/2]ip ad 10.0.0.18 30
[R2-GigabitEthernet0/2]int l0
[R2-LoopBack0]ip ad 10.1.1.2 32
---
[R3]int g0/0
[R3-GigabitEthernet0/0]ip ad 10.0.0.17 30
[R3-GigabitEthernet0/0]int g0/1
[R3-GigabitEthernet0/1]ip ad 192.168.3.254 24
[R3-GigabitEthernet0/1]int g0/2
[R3-GigabitEthernet0/2]ip ad 10.0.0.13 30
[R3-GigabitEthernet0/2]int l0
[R3-LoopBack0]ip ad 10.1.1.3 32
---
[SW1]int vlan 10
[SW1-Vlan-interface10]ip ad 192.168.1.252 24
[SW1-Vlan-interface10]int vlan 20
[SW1-Vlan-interface20]ip ad 192.168.2.252 24
[SW1-Vlan-interface20]int vlan 30
[SW1-Vlan-interface30]ip ad 10.1.2.1 30
[SW1-vlan111]int vlan 111
[SW1-Vlan-interface111]ip ad 10.0.0.6 30
[SW1-vlan111]int g1/0/1
[SW1-GigabitEthernet1/0/1]port link-type access
[SW1-GigabitEthernet1/0/1]vlan 111
[SW1-vlan111]port GigabitEthernet 1/0/1
[SW1-vlan111]int l0
[SW1-LoopBack0]ip ad 10.1.1.11 32
---
[SW2]int vlan 10
[SW2-Vlan-interface10]ip ad 192.168.1.253 24
[SW2-Vlan-interface10]int vlan 20
[SW2-Vlan-interface20]ip ad 192.168.2.253 24
[SW2-Vlan-interface20]int vlan 30
[SW2-Vlan-interface30]ip ad 10.1.2.2 30
[SW2-Vlan-interface30]int vlan 222
[SW2-Vlan-interface222]ip ad 10.0.0.10 30
[SW2]int g1/0/2
[SW2-GigabitEthernet1/0/2]port link-type access
[SW2-GigabitEthernet1/0/2]vlan 222
[SW2-vlan222]port GigabitEthernet 1/0/2
[SW2-vlan222]int l0
[SW2-LoopBack0]ip ad 10.1.1.12 32

```

3. R1 通过双线连接到互联网，配置 PPP-MP，并配置双向 chap验证

```

[R1]interface MP-group 1
[R1-MP-group1]ip ad 202.100.1.2 30
[R1-MP-group1]int s1/0
[R1-Serial1/0]ppp mp MP-group 1
[R1-Serial1/0]int s2/0

[R1-Serial2/0]ppp mp MP-group 1

```

```

[R1]local-user INTERNET class network
New local user added.
[R1-luser-network-INTERNET]password simple 123
[R1-luser-network-INTERNET]service-type ppp
[R1-luser-network-INTERNET]int s1/0
[R1-Serial1/0]ppp authentication-mode chap
[R1-Serial1/0]ppp chap user R1
[R1-Serial1/0]ppp chap password simple 456
[R1-Serial1/0]int s2/0
[R1-Serial2/0]ppp authentication-mode chap
[R1-Serial2/0]ppp chap user R1
[R1-Serial2/0]ppp chap password simple 456
---
[INTERNET]int MP-group 1
[INTERNET-MP-group1]ip ad 202.100.1.1 30
[INTERNET]int s1/0
[INTERNET-Serial1/0]ppp mp MP-group 1
[INTERNET-Serial1/0]int s2/0
[INTERNET-Serial2/0]ppp mp MP-group 1
[INTERNET]local-user R1 class network
New local user added.
[INTERNET-luser-network-R1]password simple 456
[INTERNET-luser-network-R1]service-type ppp
[INTERNET-luser-network-R1]int s1/0
[INTERNET-Serial1/0]ppp authentication-mode chap
[INTERNET-Serial1/0]ppp chap user INTERNET
[INTERNET-Serial1/0]ppp chap password simple 123
[INTERNET-Serial1/0]int s2/0
[INTERNET-Serial2/0]ppp chap user INTERNET
[INTERNET-Serial2/0]ppp authentication-mode chap
[INTERNET-Serial2/0]ppp chap password simple 123

```

4. 交换机连接 PC 的端口配置为边缘端口

```

[SW3-vlan30]int g1/0/3
[SW3-GigabitEthernet1/0/3]port access vlan 10
[SW3-GigabitEthernet1/0/3]port link-type access
[SW3-GigabitEthernet1/0/3]int g1/0/4
[SW3-GigabitEthernet1/0/3]stp edged-port
[SW3-GigabitEthernet1/0/4]port link-type access
[SW3-GigabitEthernet1/0/4]port access vlan 20
[SW3-GigabitEthernet1/0/4]stp edged-port

```

5. 在 SW1 上配置 DHCP 服务, 为 Vlan10 和 Vlan20 的 PC 动态分配 IP 地址、网关和 DNS 地址; 要求 Vlan10 的网关是 192.168.1.252, Vlan20 的网关是 192.168.2.253

```

[SW1]dhcp enable
[SW1]dhcp server ip-pool 1
[SW1-dhcp-pool-1]network 192.168.1.0 24
[SW1-dhcp-pool-1]gateway-list 192.168.1.252
[SW1-dhcp-pool-1]dns-list 1.1.1.1 2.2.2.2
[SW1-dhcp-pool-1]expired day 1
[SW1]dhcp server ip-pool 2
[SW1-dhcp-pool-2]network 192.168.2.0 24
[SW1-dhcp-pool-2]gateway-list 192.168.2.253

```

```
[SW1-dhcp-pool-2]dns-list 1.1.1.1 2.2.2.2
```

```
[SW1-dhcp-pool-2]expired day 1
```

6. 按图示分区域配置 OSPF 实现公司内部网络全网互通, ABR 的环回口宣告进骨干区域; 业务网段不允许出现协议报文

```
[R1]ospf 1
```

```
[R1-ospf-1]a 0
```

```
[R1-ospf-1-area-0.0.0.0]network 10.1.1.1 0.0.0.0
```

```
[R1-ospf-1-area-0.0.0.0]network 10.0.0.1 0.0.0.3
```

```
[R1-ospf-1-area-0.0.0.0]network 10.0.0.14 0.0.0.3
```

```
[R1-ospf-1]a 1
```

```
[R1-ospf-1-area-0.0.0.1]network 10.0.0.5 0.0.0.3
```

```
---
```

```
[R2]ospf 1
```

```
[R2-ospf-1]a 0
```

```
[R2-ospf-1-area-0.0.0.0]network 10.1.1.2 0.0.0.0
```

```
[R2-ospf-1-area-0.0.0.0]network 10.0.0.2 0.0.0.3
```

```
[R2-ospf-1-area-0.0.0.0]network 10.0.0.18 0.0.0.3
```

```
[R2-ospf-1]a 1
```

```
[R2-ospf-1-area-0.0.0.1]network 10.0.0.9 0.0.0.3
```

```
---
```

```
[R3]ospf 1
```

```
[R3-ospf-1]a 0
```

```
[R3-ospf-1-area-0.0.0.0]network 10.0.0.17 0.0.0.3
```

```
[R3-ospf-1-area-0.0.0.0]network 10.0.0.13 0.0.0.3
```

```
[R3-ospf-1-area-0.0.0.0]network 10.1.1.3 0.0.0.0
```

```
[R3-ospf-1-area-0.0.0.0]network 192.168.3.0 0.0.0.255
```

```
---
```

```
[SW1]ospf 1
```

```
[SW1-ospf-1]a 1
```

```
[SW1-ospf-1-area-0.0.0.1]network 10.1.1.11 0.0.0.0
```

```
[SW1-ospf-1-area-0.0.0.1]network 192.168.1.0 0.0.0.255
```

```
[SW1-ospf-1-area-0.0.0.1]network 192.168.2.0 0.0.0.255
```

```
[SW1-ospf-1-area-0.0.0.1]network 10.0.0.6 0.0.0.3
```

```
[SW1-ospf-1-area-0.0.0.1]network 10.1.2.1 0.0.0.3
```

```
[SW1-ospf-1]silent-interface Vlan-interface 10
```

```
[SW1-ospf-1]Nov 7 03:07:52:841 2025 SW1 OSPF/5/OSPF_NBR_CHG: OSPF 1 Neighbor  
192.168.1.253(Vlan-interface10) changed from FULL to DOWN.
```

```
[SW1-ospf-1]silent-interface Vlan-interface 20
```

```
[SW1-ospf-1]Nov 7 03:07:55:150 2025 SW1 OSPF/5/OSPF_NBR_CHG: OSPF 1 Neighbor  
192.168.2.253(Vlan-interface20) changed from FULL to DOWN.
```

```
---
```

```
[SW2-vlan222]ospf 1
```

```
[SW2-ospf-1] a 1
```

```
[SW2-ospf-1-area-0.0.0.1]network 10.1.1.12 0.0.0.0
```

```
[SW2-ospf-1-area-0.0.0.1]network 192.168.1.0 0.0.0.255
```

```
[SW2-ospf-1-area-0.0.0.1]network 192.168.2.0 0.0.0.255
```

```
[SW2-ospf-1-area-0.0.0.1]network 10.1.2.2 0.0.0.3
```

```
[SW2-ospf-1-area-0.0.0.1]network 10.0.0.10 0.0.0.3
```

```
[SW2-ospf-1]silent-interface Vlan-interface 10
```

```
[SW2-ospf-1]Nov 7 03:07:31:480 2025 SW2 OSPF/5/OSPF_NBR_CHG: OSPF 1 Neighbor  
192.168.1.252(Vlan-interface10) changed from FULL to DOWN.
```

```
silent-interface Vlan-interface 20
```

```
[SW2-ospf-1]%Nov 7 03:07:34:720 2025 SW2 OSPF/5/OSPF_NBR_CHG: OSPF 1 Neighbor
192.168.2.252(Vlan-interface20) changed from FULL to DOWN.
```

7. R1 上配置默认路由指向互联网，并引入到 OSPF

```
[R1]ip route-static 0.0.0.0 0 202.100.1.1
[R1]ospf 1
[R1-ospf-1]default-route-advertise
```

8. 配置 EASY IP，只有业务网段 192.168.1.0/24 和 192.168.2.0/24 的数据流可以通过 R1 访问互联网

```
[R1]acl basic 2000
[R1-acl-ipv4-basic-2000]rule permit source 192.168.1.0 0.0.0.255
[R1-acl-ipv4-basic-2000]rule permit source 192.168.2.0 0.0.0.255
[R1]interface MP-group 1
[R1-MP-group1]nat outbound 2000
```

9. R1 开启 TELNET 远程管理，使用用户 abc 登录，密码 abc，只允许技术部远程管理 R1

```
[R1]telnet server enable
[R1]line vty 0 4
[R1-line-vty0-4]authentication-mode scheme
[R1]local-user abc class manage
New local user added.
[R1-luser-manage-abc]password simple h3c1234567
[R1-luser-manage-abc]authorization-attribute user-role level-15
[R1-luser-manage-abc]service-type telnet
[R1]acl advanced 3000
[R1-acl-ipv4-adv-3000]rule 5 permit tcp source 192.168.2.0 0.0.0.255 destination-port eq telnet
[R1]telnet server acl 3000
```

重要配置查看

1. 接口IP地址查看

```
<R1>display ip interface brief
```

*down: administratively down

(s): spoofing (l): loopback

Interface	Physical	Protocol	IP address/Mask	VPN instance	Description
GE0/0	up	up	10.0.0.5/30	--	--
GE0/1	up	up	10.0.0.1/30	--	--
GE0/2	up	up	10.0.0.14/30	--	--
GE5/0	down	down	--	--	--
GE5/1	down	down	--	--	--
GE6/0	down	down	--	--	--
GE6/1	down	down	--	--	--
Loop0	up	up(s)	10.1.1.1/32	--	--
MP1	up	up	202.100.1.2/30	--	--
Ser1/0	up	up	--	--	--
Ser2/0	up	up	--	--	--
Ser3/0	down	down	--	--	--
Ser4/0	down	down	--	--	--

<R2>display ip interface brief

*down: administratively down

(s): spoofing (l): loopback

Interface	Physical	Protocol	IP address/Mask	VPN instance	Description
GE0/0	up	up	10.0.0.9/30	--	--
GE0/1	up	up	10.0.0.2/30	--	--
GE0/2	up	up	10.0.0.18/30	--	--
GE5/0	down	down	--	--	--
GE5/1	down	down	--	--	--
GE6/0	down	down	--	--	--
GE6/1	down	down	--	--	--
Loop0	up	up(s)	10.1.1.2/32	--	--
Ser1/0	down	down	--	--	--
Ser2/0	down	down	--	--	--
Ser3/0	down	down	--	--	--
Ser4/0	down	down	--	--	--

<R3>display ip interface brief

*down: administratively down

(s): spoofing (l): loopback

Interface	Physical	Protocol	IP address/Mask	VPN instance	Description
GE0/0	up	up	10.0.0.17/30	--	--
GE0/1	up	up	192.168.3.254/24	--	--
GE0/2	up	up	10.0.0.13/30	--	--
GE5/0	down	down	--	--	--
GE5/1	down	down	--	--	--
GE6/0	down	down	--	--	--
GE6/1	down	down	--	--	--
Loop0	up	up(s)	10.1.1.3/32	--	--
Ser1/0	down	down	--	--	--
Ser2/0	down	down	--	--	--
Ser3/0	down	down	--	--	--
Ser4/0	down	down	--	--	--

<SW1>display ip interface brief

*down: administratively down

(s): spoofing (l): loopback

Interface	Physical	Protocol	IP Address	Description
Loop0	up	up(s)	10.1.1.11	--
MGE0/0/0	down	down	--	--
Vlan10	up	up	192.168.1.252	--
Vlan20	up	up	192.168.2.252	--
Vlan30	up	up	10.1.2.1	--
Vlan111	up	up	10.0.0.6	--

<SW2>display ip interface brief

*down: administratively down

(s): spoofing (l): loopback

Interface	Physical	Protocol	IP Address	Description
Loop0	up	up(s)	10.1.1.12	--
MGE0/0/0	down	down	--	--
Vlan10	up	up	192.168.1.253	--
Vlan20	up	up	192.168.2.253	--


```

Vlan30          up      up      10.1.2.2      --
Vlan222         up      up      10.0.0.10     --
---
<SW3>display ip interface brief
*down: administratively down
(s): spoofing (l): loopback
Interface      Physical Protocol IP Address      Description
MGE0/0/0       down    down    --              --

```

```

---
<INTERNET>display ip interface brief
*down: administratively down
(s): spoofing (l): loopback
Interface      Physical Protocol IP address/Mask  VPN instance Description
GE0/0          down    down    --              --
GE0/1          down    down    --              --
GE0/2          down    down    --              --
GE5/0          down    down    --              --
GE5/1          down    down    --              --
GE6/0          down    down    --              --
GE6/1          down    down    --              --
MP1            up      up      202.100.1.1/30  --
Ser1/0         up      up      --              --
Ser2/0         up      up      --              --
Ser3/0         down    down    --              --
Ser4/0         down    down    --              --

```

2.VLAN查看

```

<SW1>display vlan brief
Brief information about all VLANs:
Supported Minimum VLAN ID: 1
Supported Maximum VLAN ID: 4094
Default VLAN ID: 1

```

VLAN ID	Name	Port
1	VLAN 0001	BAGG1 FGE1/0/53 FGE1/0/54 GE1/0/2 GE1/0/3 GE1/0/4 GE1/0/5 GE1/0/6 GE1/0/7 GE1/0/8 GE1/0/9 GE1/0/10 GE1/0/11 GE1/0/12 GE1/0/13 GE1/0/14 GE1/0/15 GE1/0/16 GE1/0/17 GE1/0/18 GE1/0/19 GE1/0/20 GE1/0/21 GE1/0/22 GE1/0/23 GE1/0/24 GE1/0/25 GE1/0/26 GE1/0/27 GE1/0/28 GE1/0/29 GE1/0/30 GE1/0/31 GE1/0/32 GE1/0/33 GE1/0/34 GE1/0/35 GE1/0/36 GE1/0/37 GE1/0/38 GE1/0/39 GE1/0/40 GE1/0/41 GE1/0/42 GE1/0/43 GE1/0/44 GE1/0/45 GE1/0/46 GE1/0/47 GE1/0/48 XGE1/0/49 XGE1/0/50 XGE1/0/51 XGE1/0/52
10	SC	BAGG1 GE1/0/2 GE1/0/3 GE1/0/4
20	JS	BAGG1 GE1/0/2 GE1/0/3 GE1/0/4

```

30      VLAN 0030      BAGG1 GE1/0/2 GE1/0/4
111     VLAN 0111     GE1/0/1
---
<SW2>display vlan brief
Brief information about all VLANs:
Supported Minimum VLAN ID: 1
Supported Maximum VLAN ID: 4094
Default VLAN ID: 1
VLAN ID   Name      Port
1         VLAN 0001  BAGG1 FGE1/0/53 FGE1/0/54
          GE1/0/1 GE1/0/3 GE1/0/4 GE1/0/5
          GE1/0/6 GE1/0/7 GE1/0/8 GE1/0/9
          GE1/0/10 GE1/0/11 GE1/0/12
          GE1/0/13 GE1/0/14 GE1/0/15
          GE1/0/16 GE1/0/17 GE1/0/18
          GE1/0/19 GE1/0/20 GE1/0/21
          GE1/0/22 GE1/0/23 GE1/0/24
          GE1/0/25 GE1/0/26 GE1/0/27
          GE1/0/28 GE1/0/29 GE1/0/30
          GE1/0/31 GE1/0/32 GE1/0/33
          GE1/0/34 GE1/0/35 GE1/0/36
          GE1/0/37 GE1/0/38 GE1/0/39
          GE1/0/40 GE1/0/41 GE1/0/42
          GE1/0/43 GE1/0/44 GE1/0/45
          GE1/0/46 GE1/0/47 GE1/0/48
          XGE1/0/49 XGE1/0/50 XGE1/0/51
          XGE1/0/52
10      SC          BAGG1 GE1/0/1 GE1/0/3 GE1/0/4
20      JS          BAGG1 GE1/0/1 GE1/0/3 GE1/0/4
30      VLAN 0030  BAGG1 GE1/0/1 GE1/0/4
222     VLAN 0222  GE1/0/2
---
<SW3>display vlan brief
Brief information about all VLANs:
Supported Minimum VLAN ID: 1
Supported Maximum VLAN ID: 4094
Default VLAN ID: 1
VLAN ID   Name      Port
1         VLAN 0001  FGE1/0/53 FGE1/0/54 GE1/0/1
          GE1/0/2 GE1/0/5 GE1/0/6 GE1/0/7
          GE1/0/8 GE1/0/9 GE1/0/10
          GE1/0/11 GE1/0/12 GE1/0/13
          GE1/0/14 GE1/0/15 GE1/0/16
          GE1/0/17 GE1/0/18 GE1/0/19
          GE1/0/20 GE1/0/21 GE1/0/22
          GE1/0/23 GE1/0/24 GE1/0/25
          GE1/0/26 GE1/0/27 GE1/0/28
          GE1/0/29 GE1/0/30 GE1/0/31
          GE1/0/32 GE1/0/33 GE1/0/34
          GE1/0/35 GE1/0/36 GE1/0/37
          GE1/0/38 GE1/0/39 GE1/0/40
          GE1/0/41 GE1/0/42 GE1/0/43
          GE1/0/44 GE1/0/45 GE1/0/46

```

```
GE1/0/47 GE1/0/48 XGE1/0/49
XGE1/0/50 XGE1/0/51 XGE1/0/52
10      SC      GE1/0/1 GE1/0/2 GE1/0/3
20      JS      GE1/0/1 GE1/0/2 GE1/0/4
30      VLAN 0030
```

3.DHCP查看

```
<SW1>display dhcp server ip-in-use
```

IP address	Client identifier/ Hardware address	Lease expiration	Type
192.168.1.1	0032-3661-352e-3735- 6462-2e30-3830-362d- 4745-302f-302f-31	Nov 9 01:31:58 2025	Auto(C)
192.168.2.1	0032-3661-352e-3736- 6633-2e30-3930-362d- 4745-302f-302f-31	Nov 9 01:32:06 2025	Auto(C)

```
<SW1>display dhcp server statistics
```

```
Pool number: 2
Pool utilization: 0.78%
Bindings:
  Automatic: 2
  Manual: 0
  Expired: 0
Conflict: 0
Messages received: 4
  DHCPDISCOVER: 2
  DHCPREQUEST: 2
  DHCPDECLINE: 0
  DHCPRELEASE: 0
  DHCPINFORM: 0
  BOOTPREQUEST: 0
Messages sent: 4
  DHCPOFFER: 2
  DHCPACK: 2
  DHCPNAK: 0
  BOOTPREPLY: 0
Bad Messages: 0
```

```
<SW1>display dhcp server pool
```

```
Pool name: 1
```

```
Network: 192.168.1.0 mask 255.255.255.0
dns-list 1.1.1.1 2.2.2.2
expired day 1 hour 0 minute 0 second 0
reserve expired-ip enable
reserve expired-ip mode client-id time 4294967295 limit 256000
gateway-list 192.168.1.252
```

```
Pool name: 2
```

```
Network: 192.168.2.0 mask 255.255.255.0
dns-list 1.1.1.1 2.2.2.2
expired day 1 hour 0 minute 0 second 0
reserve expired-ip enable
reserve expired-ip mode client-id time 4294967295 limit 256000
gateway-list 192.168.2.253
```

4.OSPF邻居

```
<R1>display ospf peer
```

```
OSPF Process 1 with Router ID 10.1.1.1
Neighbor Brief Information
```

```
Area: 0.0.0.0
```

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.2	10.0.0.2	1	34	Full/DR	GE0/1
10.1.1.3	10.0.0.13	1	34	Full/DR	GE0/2

```
Area: 0.0.0.1
```

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.11	10.0.0.6	1	37	Full/DR	GE0/0

```
---
```

```
<R2>display ospf peer
```

```
OSPF Process 1 with Router ID 10.1.1.2
Neighbor Brief Information
```

```
Area: 0.0.0.0
```

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.1	10.0.0.1	1	30	Full/BDR	GE0/1
10.1.1.3	10.0.0.17	1	38	Full/DR	GE0/2

```
Area: 0.0.0.1
```

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.12	10.0.0.10	1	33	Full/BDR	GE0/0

```
---
```

```
[R3-ospf-1-area-0.0.0.0] display ospf peer
```

```
OSPF Process 1 with Router ID 10.1.1.3
Neighbor Brief Information
```

```
Area: 0.0.0.0
```

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.2	10.0.0.18	1	38	Full/BDR	GE0/0
10.1.1.1	10.0.0.14	1	38	Full/BDR	GE0/2

```
---
```

```
<SW1>display ospf peer
```

```
OSPF Process 1 with Router ID 10.1.1.11
Neighbor Brief Information
```

```
Area: 0.0.0.1
```

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.12	10.1.2.2	1	32	Full/DR	Vlan30
10.1.1.1	10.0.0.5	1	30	Full/BDR	Vlan111

```
---
```

```
<SW2>display ospf peer
```

```
OSPF Process 1 with Router ID 10.1.1.12
```

```
Neighbor Brief Information
```

Area: 0.0.0.1

Router ID	Address	Pri	Dead-Time	State	Interface
10.1.1.11	10.1.2.1	1	40	Full/BDR	Vlan30
10.1.1.2	10.0.0.9	1	39	Full/DR	Vlan222

5.NAT查看

<R1>display nat all

NAT outbound information:

Totally 1 NAT outbound rules.

Interface: MP-group1

ACL: 2000

Address group ID: ---

Port-preserved: N NO-PAT: N Reversible: N

NAT counting: 0

Config status: Active

NAT logging:

Log enable : Disabled

Flow-begin : Disabled

Flow-end : Disabled

Flow-active : Disabled

Port-block-assign : Disabled

Port-block-withdraw : Disabled

Alarm : Disabled

NO-PAT IP usage : Disabled

NAT mapping behavior:

Mapping mode : Address and Port-Dependent

ACL : ---

Config status: Active

NAT ALG:

DNS : Enabled

FTP : Enabled

H323 : Disabled

ICMP-ERROR : Enabled

ILS : Disabled

MGCP : Disabled

NBT : Disabled

PPTP : Enabled

RTSP : Enabled

RSH : Disabled

SCCP : Disabled

SIP : Disabled

SQLNET : Disabled

TFTP : Disabled

XDMCP : Disabled

Static NAT load balancing: Disabled

NAT link-switch recreate-session: Disabled

6. IP路由表

[R1-acl-ipv4-adv-3000]dis ip routing-table

Destinations : 31 Routes : 32

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/0	Static	60	0	202.100.1.1	MP1
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.0/30	Direct	0	0	10.0.0.1	GE0/1
10.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.3/32	Direct	0	0	10.0.0.1	GE0/1
10.0.0.4/30	Direct	0	0	10.0.0.5	GE0/0
10.0.0.5/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.7/32	Direct	0	0	10.0.0.5	GE0/0
10.0.0.8/30	O_INTRA	10	3	10.0.0.6	GE0/0
10.0.0.12/30	Direct	0	0	10.0.0.14	GE0/2
10.0.0.14/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.15/32	Direct	0	0	10.0.0.14	GE0/2
10.0.0.16/30	O_INTRA	10	2	10.0.0.2	GE0/1
	O_INTRA	10	2	10.0.0.13	GE0/2
10.1.1.1/32	Direct	0	0	127.0.0.1	InLoop0
10.1.1.3/32	O_INTRA	10	1	10.0.0.13	GE0/2
10.1.1.11/32	O_INTRA	10	1	10.0.0.6	GE0/0
10.1.1.12/32	O_INTRA	10	2	10.0.0.6	GE0/0
10.1.2.0/30	O_INTRA	10	2	10.0.0.6	GE0/0
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.0/24	O_INTRA	10	2	10.0.0.6	GE0/0
192.168.2.0/24	O_INTRA	10	2	10.0.0.6	GE0/0
192.168.3.0/24	O_INTRA	10	2	10.0.0.13	GE0/2
202.100.1.0/30	Direct	0	0	202.100.1.2	MP1
202.100.1.1/32	Direct	0	0	202.100.1.1	MP1
202.100.1.2/32	Direct	0	0	127.0.0.1	InLoop0
202.100.1.3/32	Direct	0	0	202.100.1.2	MP1
224.0.0.0/4	Direct	0	0	0.0.0.0	NULL0
224.0.0.0/24	Direct	0	0	0.0.0.0	NULL0
255.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0

<R2>dis ip routing-table

Destinations : 28 Routes : 29

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/0	O_ASE2	150	1	10.0.0.1	GE0/1
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.0/30	Direct	0	0	10.0.0.2	GE0/1
10.0.0.2/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.3/32	Direct	0	0	10.0.0.2	GE0/1
10.0.0.4/30	O_INTRA	10	3	10.0.0.10	GE0/0
10.0.0.8/30	Direct	0	0	10.0.0.9	GE0/0
10.0.0.9/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.11/32	Direct	0	0	10.0.0.9	GE0/0

10.0.0.12/30	O_INTRA	10	2	10.0.0.1	GE0/1
	O_INTRA	10	2	10.0.0.17	GE0/2
10.0.0.16/30	Direct	0	0	10.0.0.18	GE0/2
10.0.0.18/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.19/32	Direct	0	0	10.0.0.18	GE0/2
10.1.1.1/32	O_INTRA	10	1	10.0.0.1	GE0/1
10.1.1.2/32	Direct	0	0	127.0.0.1	InLoop0
10.1.1.3/32	O_INTRA	10	1	10.0.0.17	GE0/2
10.1.1.11/32	O_INTRA	10	2	10.0.0.10	GE0/0
10.1.1.12/32	O_INTRA	10	1	10.0.0.10	GE0/0
10.1.2.0/30	O_INTRA	10	2	10.0.0.10	GE0/0
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.0/24	O_INTRA	10	2	10.0.0.10	GE0/0
192.168.2.0/24	O_INTRA	10	2	10.0.0.10	GE0/0
192.168.3.0/24	O_INTRA	10	2	10.0.0.17	GE0/2
224.0.0.0/4	Direct	0	0	0.0.0.0	NULL0
224.0.0.0/24	Direct	0	0	0.0.0.0	NULL0
255.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0

<R3>dis ip routing-table

Destinations : 27 Routes : 31

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/0	O_ASE2	150	1	10.0.0.14	GE0/2
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.0/30	O_INTRA	10	2	10.0.0.14	GE0/2
	O_INTRA	10	2	10.0.0.18	GE0/0
10.0.0.4/30	O_INTER	10	2	10.0.0.14	GE0/2
10.0.0.8/30	O_INTER	10	2	10.0.0.18	GE0/0
10.0.0.12/30	Direct	0	0	10.0.0.13	GE0/2
10.0.0.13/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.15/32	Direct	0	0	10.0.0.13	GE0/2
10.0.0.16/30	Direct	0	0	10.0.0.17	GE0/0
10.0.0.17/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.19/32	Direct	0	0	10.0.0.17	GE0/0
10.1.1.1/32	O_INTRA	10	1	10.0.0.14	GE0/2
10.1.1.3/32	Direct	0	0	127.0.0.1	InLoop0
10.1.1.11/32	O_INTER	10	2	10.0.0.14	GE0/2
10.1.1.12/32	O_INTER	10	2	10.0.0.18	GE0/0
10.1.2.0/30	O_INTER	10	3	10.0.0.14	GE0/2
	O_INTER	10	3	10.0.0.18	GE0/0
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.0/24	O_INTER	10	3	10.0.0.14	GE0/2
	O_INTER	10	3	10.0.0.18	GE0/0
192.168.2.0/24	O_INTER	10	3	10.0.0.14	GE0/2
	O_INTER	10	3	10.0.0.18	GE0/0
192.168.3.0/24	Direct	0	0	192.168.3.254	GE0/1
192.168.3.254/32	Direct	0	0	127.0.0.1	InLoop0


```

192.168.3.255/32 Direct 0 0 192.168.3.254 GE0/1
224.0.0.0/4 Direct 0 0 0.0.0.0 NULL0
224.0.0.0/24 Direct 0 0 0.0.0.0 NULL0
255.255.255.255/32 Direct 0 0 127.0.0.1 InLoop0

```

<SW1>dis ip routing-table

Destinations : 34 Routes : 35

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/0	O_ASE2	150	1	10.0.0.5	Vlan111
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.0/30	O_INTER	10	2	10.0.0.5	Vlan111
10.0.0.4/30	Direct	0	0	10.0.0.6	Vlan111
10.0.0.4/32	Direct	0	0	10.0.0.6	Vlan111
10.0.0.6/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.7/32	Direct	0	0	10.0.0.6	Vlan111
10.0.0.8/30	O_INTRA	10	2	10.1.2.2	Vlan30
10.0.0.12/30	O_INTER	10	2	10.0.0.5	Vlan111
10.0.0.16/30	O_INTER	10	3	10.0.0.5	Vlan111
				10.1.2.2	Vlan30
10.1.1.1/32	O_INTER	10	1	10.0.0.5	Vlan111
10.1.1.3/32	O_INTER	10	2	10.0.0.5	Vlan111
10.1.1.11/32	Direct	0	0	127.0.0.1	InLoop0
10.1.1.12/32	O_INTRA	10	1	10.1.2.2	Vlan30
10.1.2.0/30	Direct	0	0	10.1.2.1	Vlan30
10.1.2.0/32	Direct	0	0	10.1.2.1	Vlan30
10.1.2.1/32	Direct	0	0	127.0.0.1	InLoop0
10.1.2.3/32	Direct	0	0	10.1.2.1	Vlan30
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.0/24	Direct	0	0	192.168.1.252	Vlan10
192.168.1.0/32	Direct	0	0	192.168.1.252	Vlan10
192.168.1.252/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.255/32	Direct	0	0	192.168.1.252	Vlan10
192.168.2.0/24	Direct	0	0	192.168.2.252	Vlan20
192.168.2.0/32	Direct	0	0	192.168.2.252	Vlan20
192.168.2.252/32	Direct	0	0	127.0.0.1	InLoop0
192.168.2.255/32	Direct	0	0	192.168.2.252	Vlan20
192.168.3.0/24	O_INTER	10	3	10.0.0.5	Vlan111
224.0.0.0/4	Direct	0	0	0.0.0.0	NULL0
224.0.0.0/24	Direct	0	0	0.0.0.0	NULL0
255.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0

<SW2>dis ip routing-table

Destinations : 34 Routes : 36

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/0	O_ASE2	150	1	10.1.2.1	Vlan30
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0

10.0.0.0/30	O_INTER	10	2	10.0.0.9	Vlan222
10.0.0.4/30	O_INTRA	10	2	10.1.2.1	Vlan30
10.0.0.8/30	Direct	0	0	10.0.0.10	Vlan222
10.0.0.8/32	Direct	0	0	10.0.0.10	Vlan222
10.0.0.10/32	Direct	0	0	127.0.0.1	InLoop0
10.0.0.11/32	Direct	0	0	10.0.0.10	Vlan222
10.0.0.12/30	O_INTER	10	3	10.0.0.9	Vlan222
				10.1.2.1	Vlan30
10.0.0.16/30	O_INTER	10	2	10.0.0.9	Vlan222
10.1.1.1/32	O_INTER	10	2	10.0.0.9	Vlan222
				10.1.2.1	Vlan30
10.1.1.3/32	O_INTER	10	2	10.0.0.9	Vlan222
10.1.1.11/32	O_INTRA	10	1	10.1.2.1	Vlan30
10.1.1.12/32	Direct	0	0	127.0.0.1	InLoop0
10.1.2.0/30	Direct	0	0	10.1.2.2	Vlan30
10.1.2.0/32	Direct	0	0	10.1.2.2	Vlan30
10.1.2.2/32	Direct	0	0	127.0.0.1	InLoop0
10.1.2.3/32	Direct	0	0	10.1.2.2	Vlan30
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.0/24	Direct	0	0	192.168.1.253	Vlan10
192.168.1.0/32	Direct	0	0	192.168.1.253	Vlan10
192.168.1.253/32	Direct	0	0	127.0.0.1	InLoop0
192.168.1.255/32	Direct	0	0	192.168.1.253	Vlan10
192.168.2.0/24	Direct	0	0	192.168.2.253	Vlan20
192.168.2.0/32	Direct	0	0	192.168.2.253	Vlan20
192.168.2.253/32	Direct	0	0	127.0.0.1	InLoop0
192.168.2.255/32	Direct	0	0	192.168.2.253	Vlan20
192.168.3.0/24	O_INTER	10	3	10.0.0.9	Vlan222
224.0.0.0/4	Direct	0	0	0.0.0.0	NULL0
224.0.0.0/24	Direct	0	0	0.0.0.0	NULL0
255.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0

<SW3>dis ip routing-table

Destinations : 8 Routes : 8

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
224.0.0.0/4	Direct	0	0	0.0.0.0	NULL0
224.0.0.0/24	Direct	0	0	0.0.0.0	NULL0
255.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0

<INTERNET>dis ip routing-table

Destinations : 11 Routes : 11

Destination/Mask	Proto	Pre	Cost	NextHop	Interface
0.0.0.0/32	Direct	0	0	127.0.0.1	InLoop0
127.0.0.0/8	Direct	0	0	127.0.0.1	InLoop0
127.0.0.1/32	Direct	0	0	127.0.0.1	InLoop0
127.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0
202.100.1.0/30	Direct	0	0	202.100.1.1	MP1
202.100.1.1/32	Direct	0	0	127.0.0.1	InLoop0
202.100.1.2/32	Direct	0	0	202.100.1.2	MP1
202.100.1.3/32	Direct	0	0	202.100.1.1	MP1
224.0.0.0/4	Direct	0	0	0.0.0.0	NULL0
224.0.0.0/24	Direct	0	0	0.0.0.0	NULL0
255.255.255.255/32	Direct	0	0	127.0.0.1	InLoop0

7.连通性与Telnet测试

<H3C>ping 202.100.1.1

Ping 202.100.1.1 (202.100.1.1): 56 data bytes, press CTRL_C to break
56 bytes from 202.100.1.1: icmp_seq=0 ttl=253 time=6.619 ms
56 bytes from 202.100.1.1: icmp_seq=1 ttl=253 time=3.215 ms

<H3C>ping 202.100.1.2

Ping 202.100.1.2 (202.100.1.2): 56 data bytes, press CTRL_C to break
56 bytes from 202.100.1.2: icmp_seq=0 ttl=254 time=1.413 ms
56 bytes from 202.100.1.2: icmp_seq=1 ttl=254 time=1.264 ms

<H3C>ping 192.168.2.1

Ping 192.168.2.1 (192.168.2.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.1: icmp_seq=0 ttl=254 time=2.729 ms
56 bytes from 192.168.2.1: icmp_seq=1 ttl=254 time=1.955 ms

<H3C>ping 192.168.3.1

Ping 192.168.3.1 (192.168.3.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.3.1: icmp_seq=0 ttl=252 time=2.275 ms
56 bytes from 192.168.3.1: icmp_seq=1 ttl=252 time=2.280 ms

<H3C>telnet 202.100.1.2

Trying 202.100.1.2 ...
Press CTRL+K to abort
Connected to 202.100.1.2 ...
Failed to connect to the remote host!

<H3C>ping 202.100.1.1

Ping 202.100.1.1 (202.100.1.1): 56 data bytes, press CTRL_C to break
56 bytes from 202.100.1.1: icmp_seq=0 ttl=253 time=1.668 ms
56 bytes from 202.100.1.1: icmp_seq=1 ttl=253 time=2.363 ms

<H3C>ping 202.100.1.2

Ping 202.100.1.2 (202.100.1.2): 56 data bytes, press CTRL_C to break
56 bytes from 202.100.1.2: icmp_seq=0 ttl=254 time=1.575 ms
56 bytes from 202.100.1.2: icmp_seq=1 ttl=254 time=1.825 ms

<H3C>ping 192.168.1.1

Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break

```
56 bytes from 192.168.1.1: icmp_seq=0 ttl=254 time=1.903 ms
56 bytes from 192.168.1.1: icmp_seq=1 ttl=254 time=1.608 ms
```

```
<H3C>ping 192.168.3.1
```

```
Ping 192.168.3.1 (192.168.3.1): 56 data bytes, press CTRL_C to break
```

```
56 bytes from 192.168.3.1: icmp_seq=0 ttl=252 time=2.706 ms
```

```
56 bytes from 192.168.3.1: icmp_seq=1 ttl=252 time=2.008 ms
```

```
<H3C>telnet 202.100.1.2
```

```
Trying 202.100.1.2 ...
```

```
Press CTRL+K to abort
```

```
Connected to 202.100.1.2 ...
```

```
*****
* Copyright (c) 2004-2021 New H3C Technologies Co., Ltd. All rights reserved.*
* Without the owner's prior written consent,                               *
* no decompiling or reverse-engineering shall be allowed.                 *
*****
```

```
Login: abc
```

```
Password:
```

```
<R1>dir
```

```
Directory of flash: (VFAT)
```

0	drw-	-	Nov 07 2025 01:47:02	diagfile
1	-rw-	286	Nov 07 2025 03:37:54	ifindex.dat
2	-rw-	43136	Nov 07 2025 01:47:02	licbackup
3	-rw-	43136	Nov 07 2025 01:47:02	licnormal
4	drw-	-	Nov 07 2025 01:47:02	logfile
5	-rw-	0	Nov 07 2025 01:47:02	msr36-cmw710-boot-r0424p22.bin
6	-rw-	0	Nov 07 2025 01:47:02	msr36-cmw710-system-r0424p22.bin
7	drw-	-	Nov 07 2025 01:47:02	seclog
8	-rw-	3717	Nov 07 2025 03:37:54	startup.cfg
9	-rw-	55116	Nov 07 2025 03:37:54	startup.mdb

```
1046512 KB total (1046340 KB free)
```

```
---
```

```
<H3C>ping 202.100.1.1
```

```
Ping 202.100.1.1 (202.100.1.1): 56 data bytes, press CTRL_C to break
```

```
Request time out
```

```
Request time out
```

```
<H3C>ping 202.100.1.2
```

```
Ping 202.100.1.2 (202.100.1.2): 56 data bytes, press CTRL_C to break
```

```
56 bytes from 202.100.1.2: icmp_seq=0 ttl=254 time=1.050 ms
```

```
56 bytes from 202.100.1.2: icmp_seq=1 ttl=254 time=0.885 ms
```

```
<H3C>ping 192.168.1.1
```

```
Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break
```

```
56 bytes from 192.168.1.1: icmp_seq=0 ttl=252 time=2.541 ms
```

```
56 bytes from 192.168.1.1: icmp_seq=1 ttl=252 time=2.061 ms
```

```
<H3C>ping 192.168.2.1
Ping 192.168.2.1 (192.168.2.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.1: icmp_seq=0 ttl=252 time=3.205 ms
56 bytes from 192.168.2.1: icmp_seq=1 ttl=252 time=2.244 ms

<H3C>telnet 202.100.1.2
Trying 202.100.1.2 ...
Press CTRL+K to abort
Connected to 202.100.1.2 ...
Failed to connect to the remote host!
=====

8.acl查看
[R1-acl-ipv4-adv-3000]dis acl all
Basic IPv4 ACL 2000, 2 rules,
ACL's step is 5
  rule 0 permit source 192.168.1.0 0.0.0.255 (2 times matched)
  rule 5 permit source 192.168.2.0 0.0.0.255 (1 times matched)

Advanced IPv4 ACL 3000, 1 rule,
ACL's step is 5
  rule 5 permit tcp source 192.168.2.0 0.0.0.255 destination-port eq telnet (2 times matched)
```

附录

在其中的实验中遇见的问题及解决方法如下:

1. OSPF的建立出现了问题, SW1和SW2和各自的上行路由器之间无法建立OSPF邻居连接;
 - a. 使用了ping对端接口IP地址发现无法ping通
 - b. 在SW1、SW2上使用dis ip in br查看接口IP地址时, 发现物理端口和协议双Down
2. 在查询资料后, 得到两种解决方式:
 - a. 将端口划入到VLAN中 (即VLAN111和VLAN222)
 - b. 在接口上开启三层转发功能, 再在接口上配置IP地址
3. 在尝试了第二种方法后发现交换机支持不了, 所以采用了第一种方法将上行与路由器连接的物理端口划入至VLAN中实现了端口与协议的双UP, 成功将OSPF的邻居建立起来